

(KTXFI1EBNF) Physics I. Test 2

May 18, 2026

1. Estimate the average depth of the Indian Ocean if a tsunami takes 24 hours to travel from Jakarta to Mogadishu. (Note: The propagation speed of shallow-water waves is $v = \sqrt{gh}$. The distance is 17000 km. $g = 9.81 \text{ m/s}^2$.)

(10 points)

2. A projectile is launched at an angle of 45° with an initial velocity of 50 m/s ($g = 9.81 \text{ m/s}^2$). Calculate:

- (a) The time to reach the peak of the trajectory.
- (b) The maximum height.
- (c) The horizontal range (impact distance).
- (d) The total time of flight.

(10 points)

3. We place a copper piece with a mass of 100 gramms and a temperature of 40°C into 500 gramms of water at 80°C . What will be the common temperature? The specific heat capacity of copper is $3.85 \cdot 10^2 \frac{\text{J}}{\text{kg K}}$ and the specific heat capacity of water is $4.19 \cdot 10^3 \frac{\text{J}}{\text{kg K}}$.

(10 points)

4. What is the entropy change if the volume of $m = 14 \text{ g}$ of nitrogen changes from $V_1 = 30 \text{ l}$ to $V_2 = 120 \text{ l}$, and its temperature changes from $T_1 = 27^\circ\text{C}$ to $T_2 = 327^\circ\text{C}$? The molar mass is $M = 28 \text{ g/mol}$, $R = 8.31 \frac{\text{J}}{\text{mol K}}$.

(10 points)

5. Determine the acceleration of a proton ($m = 1.67 \cdot 10^{-27} \text{ kg}$, $e = 1.6 \cdot 10^{-19} \text{ C}$) in an electric field of strength $E = 10000 \text{ N/C}$. How many times greater is this acceleration than the acceleration due to gravity?

(10 points)

Requirement for obtaining the course signature: achieving at least 40% (20 points) on the midterm exam.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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